# Ahmed Jawad Rashid

Curriculum Vitae

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## Personal Information


Junior Lecturer
Department of Electrical and Electronic Engineering
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## Research Interest

Computer Vision, Edge AI, Reinforcement Learning

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## Education

**Master of Science (M.Sc)** in Electrical and Electronic Engineering — *Sep 2024 – Nov 2026*
**Islamic University of Technology**, Dhaka, Bangladesh
**CGPA**: 4.00/4.00; (Supervisor: Prof. Dr. Golam Sarowar)

**Bachelor of Science (B.Sc)** in Electrical and Electronic Engineering — *Jan 2020 – June 2024*
**Islamic University of Technology**, Dhaka, Bangladesh
**CGPA**: 3.92/4.00; 98 Percentile in the class of 2025 (Supervisor: Mr. Nadim Ahmed)

**Higher Secondary School Certificate** — *Aug 2017 – May 2019*
**Notre Dame College**, Dhaka, Bangladesh
**GPA**: 5.00/5.00

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## Research Work

### Dissertation

**Rashid, A. J.**, 2026. Printed Circuit Board Defect Detection Using Convolutional Neural Networks on Multi-Source Physically Fabricated Datasets (M.Sc. dissertation, Department of Electrical and Electronic Engineering (EEE), Islamic University of Technology (IUT), Board Bazar, Gazipur-1704, Bangladesh). [PDF]

### Journal

**A. J. Rashid**, N. Ahmed, and G. Sarowar, "P2LK-YOLO: Reallocating feature resolution and large-kernel context for oriented detection of printed circuit board surface defects," submitted to *Results in Engineering*, 2026. [PDF]

**A. J. Rashid**, M. A. Ullah, A. Isfara, N. Ahmed, M. M. Mian, and M. M. Shalehin, "PCB-Defect: An annotated dataset for surface defect detection in printed circuit boards," *Data in Brief*, vol. 64, Art. no. 112296, 2026. [PDF]

### Dataset

Azam, Ashraful; Ishti, Masum Mushfiq; Shams, Khondakar Muntaseer; **Rashid, Ahmed Jawad**; Ahmed, Nadim (2026), "DefectDet: An Annotated Dataset of Printed Circuit Boards with Surface Defects", Mendeley Data, V1, doi: 10.17632/t9d9zs3bmb.1

**Rashid, Ahmed Jawad**; Ullah, Mohammad Aman; Isfara, Adiba; Ahmed, Nadim; Mian, Mamun; Shalehin, Md. Mashur (2025), "PCB-Defect: An Annotated Dataset for Surface Defect Detection in Printed Circuit Boards", Mendeley Data, V1, doi: 10.17632/vdj74sngvn.2

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## Teaching Experience

**Junior Lecturer** — *Sep 2024 – Present*
*Islamic University of Technology*, Dhaka, Bangladesh

- EEE 4709: Artificial Intelligence and Machine Learning
- EEE 4308: Digital Logic Design Lab
- EEE 4404: Analog Communication Lab
- EEE 4542: Wireless Communication Lab
- EEE 4616: Microprocessor and Assembly Language Lab
- EEE 4704: Digital Communication Lab

— EEE 4710: Artificial Intelligence and Machine Learning Lab

## Technical Skills

**Languages**: Python, C/C++, MATLAB, Assembly
**Deep Learning & ML**: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face
**Computer Vision**: OpenCV, Ultralytics, Detectron2, torchvision, MMDetection
**Reinforcement Learning**: Gymnasium, Stable-Baselines3 (PPO)
**Data & Scientific Computing**: NumPy, pandas, SciPy, Matplotlib, Seaborn, Plotly
**Electronics & Embedded Systems**: Arduino, Raspberry Pi
**Software**: KiCAD, Atoll, Proteus
**Developer Tools**: Git, GitHub, Linux, TensorBoard

## Projects

**PCBKIT: PCB Defect Dataset Toolkit** [Link]
*Python, OpenCV, Shapely, Oriented Object Detection*
— Built an open-source toolkit that tiles high-resolution PCB images without cutting any defect across tile boundaries
— Added a verified YOLO-OBB to DOTA annotation converter, so one dataset trains both Ultralytics and MMRotate models

**Traffic Vehicle Detection and Counting** [Link]
*Python, YOLOv8, OpenCV, Multi-Object Tracking*
— Trained YOLOv8 on a custom car and truck dataset (0.93 mAP50) and added tracking to count vehicles crossing a line in traffic video

**Digital Snake and Ladder Game** [Link]
*Digital Logic, 74-Series ICs, Proteus*
— Built a two-player game from logic ICs on a 4×4 LED board, with circuits for dice rolls, moves, snakes, ladders and win detection

**SAP-1 8-bit Computer** [Link]
*Computer Architecture, Bus Design, Proteus*
— Designed and simulated an 8-bit computer with a shared bus and ring-counter controller that runs LDA, ADD, SUB, OUT and HLT

**Bluetooth Interfacing with 8051 Microcontroller** [Link]
*8051 Assembly, HC-05, Serial Communication*
— Programmed Bluetooth-controlled features in assembly, including a countdown timer, Morse code, LED matrix display, encryption and relay switching

**Ball Balancing Platform Control** [Link]
*PID Control, Arduino, Servo Motor, Ultrasonic Sensing*
— Built an Arduino ball-on-beam balancer that uses ultrasonic feedback and a servo, with a tuned PID controller holding the ball at a set position

## Awards and Recognition

**2nd/100**, Transparency Intl. Bangladesh's Idea Contest on Renewable Energy (Mentor), 2024
**Top 10 out of 1000+**, Huawei Seeds for the Future, Bangladesh 2023, 2023
**2nd/30**, Hult Prize at IUT 2023, 2023
**National Finalist**, EWU presents Analytics 2.0, 2021
**Global Finalist**, Int. Rover Design Challenge (Technical Documentation Team, IUT Mars Rover), 2021

## Other Experiences

**Vice President** *2023 – 2024*
*IUT Career and Business Society*, Dhaka, Bangladesh

**Senior Electrical Team Member** *2022 – 2023*
*IUT Mars Rover*, Dhaka, Bangladesh

**President** *2018 – 2019*
*Notre Dame International Understanding and Relation Club*, Dhaka, Bangladesh